

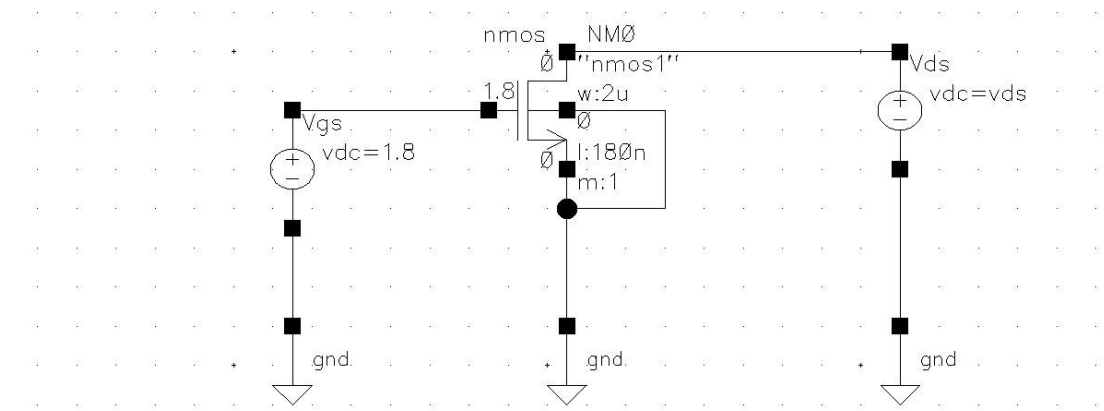
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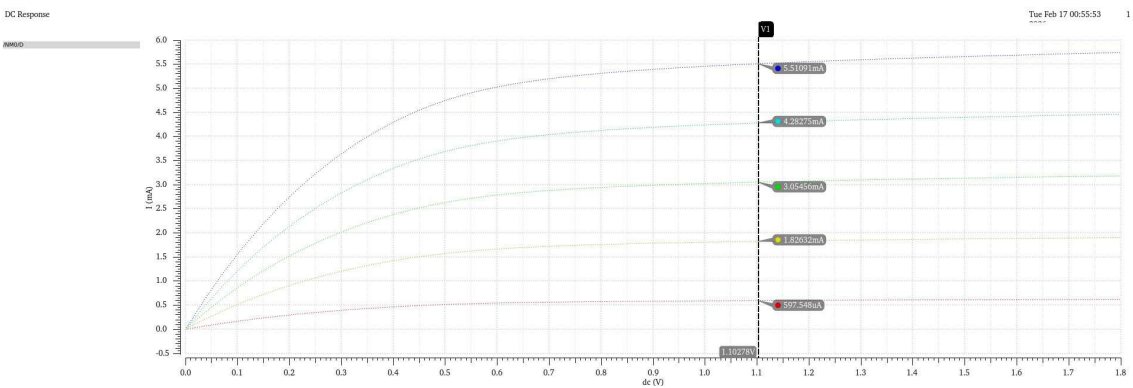
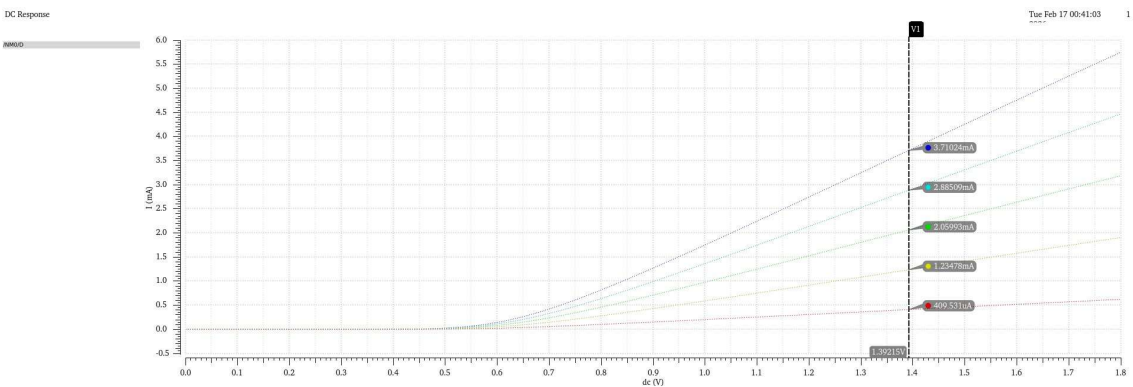
LAB EXPERIMENT 1

Aim: Extracting small signal parameters of 180nm NMOS Transistor.

Circuit Diagram:



Results:



	gm	gmb	gmb	gm	vdsat	vdsat	vbi
0	0	0	0	0	0	0	0
0	-1.47792e1	3.35674e1	-1.00311	1.36932e1	1.36932e1	2.36243e1	0
0	-4.82667e1	12.9918e1	-3.32847e1	3.91615e1	3.91615e1	7.10482e1	0
0	-6.17922e1	22.0248e1	-5.87032e1	6.46279e1	6.46279e1	1.14362e2	0
0	-11.5244e1	31.0578e1	-8.03921e1	9.03941e1	9.03941e1	1.67668e2	0
0	-14.8732e1	40.0911e1	-1.03446e2	1.15581e2	1.15581e2	2.15013e2	0

Conclusion: The small-signal parameters of the 180 nm NMOS transistor, such as transconductance (gm), output resistance (ro), and capacitances, were successfully extracted from DC operating point analysis. The results show that gm increases with drain current, while ro decreases due to channel length modulation. These parameters are essential for accurate analog circuit design and gain estimation in 180 nm CMOS technology.